OSS4008A – Project Progress 2 Update

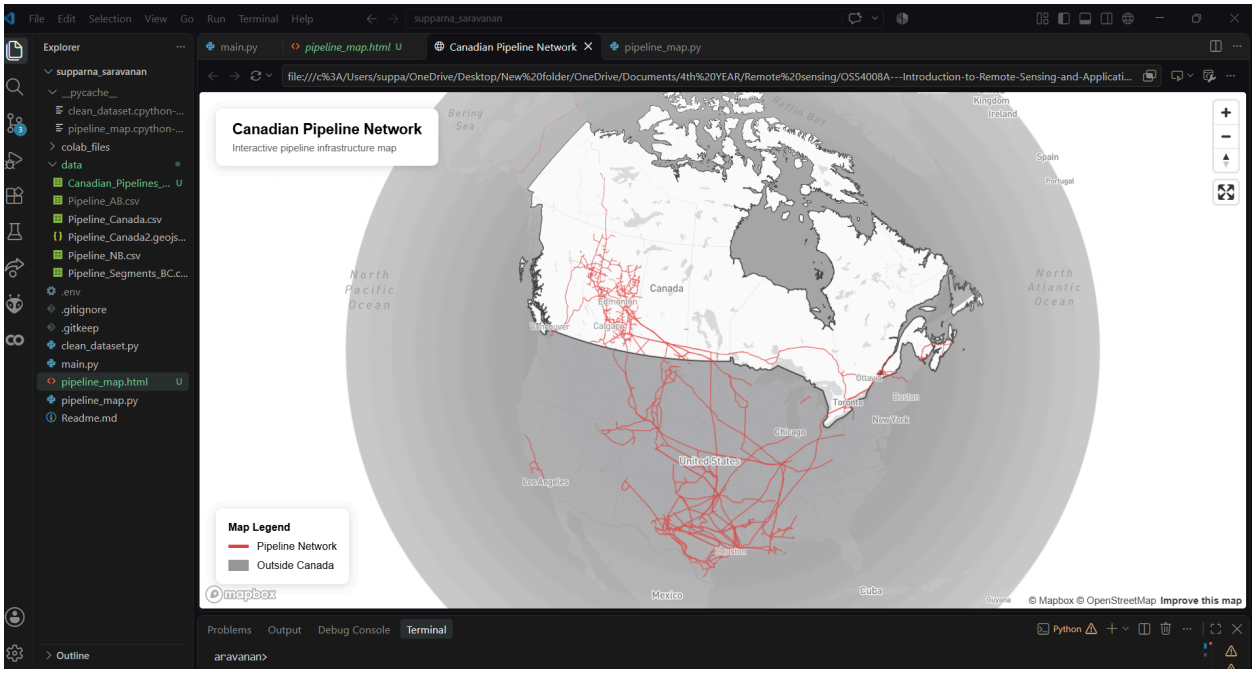
Project: Mapping the Oil Industry and Pipelines in Canada
Date: November 12, 2026

1. Work Completed

- Researched, sourced, and organized geographical datasets related to Canada's oil industry and pipelines from multiple provinces.
- Developed code to clean, standardize, and merge the provincial datasets into one consolidated database.
- Developed Python code to create an interactive map using the Mapbox API.
- Integrated the database into the map and added functionality to display specific data sections while keeping the website responsive.

2. Map and UI Development

The map is completed and displays the selected datasets. I have not yet developed a separate dashboard or user interface. My next step is to connect the map to a simple interface with buttons for selecting oil industry and pipeline data.



3. Planned UI Features

- **Oil Industry:** Display oil industry locations.
- **Pipelines:** Display pipeline data.

- **Map View:** View the data on the map.
- **Search:** Find a specific location.
- **Reset:** Return to the default map view.

4. AI Integration

I have not started the AI component yet. I plan to research suitable AI methods and explore how they could be used to identify oil-related infrastructure and connect the results to the map.

5. Next Steps

Before Progress 3, I plan to:

- Develop the basic user interface.
- Add buttons to control the map layers.
- Connect the interface to my existing Python code.
- Research possible AI methods.
- Test the map and its features.